

EEG based BCI for Seamless Home Automation

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Abstract—Brain-computer interface (BCI) facilitates direct interaction between the human brain and external devices. This research investigates a deep learning-based method for motor imagery classification from electroencephalography (EEG) signals. In contrast to existing works that utilize a greater number of electrodes, this research utilizes a smaller set of electrodes (Cz, C3, C4, F3, Fz, F4, P3, P4) to reduce hardware requirements while preserving classification accuracy. Signal preprocessing comprises a bandpass filter (8–30 Hz) that is realized by a Butterworth filter. Spectral transformation is carried out by continuous wavelet transform (CWT) with a complex Morlet wavelet. A convolutional neural network (CNN) is architected with three convolutional layers, batch normalization, and dropout regularization for avoiding overfitting. The model is trained with categorical cross-entropy loss with Adam optimization. Experimental results indicate that our approach attains competitive classification accuracy with lower computational complexity and hardware demands than current methods.

Keywords : Brain Computer Interface, IOT, Automation, Deep learning, EEG, CNN, Biomedical.

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) have proven to be a ground breaking technology that permits the direct interaction between the human brain and external devices. Through the interpretation of neural activity, BCIs offer an interface through which people can operate devices without actual physical interaction, making them highly beneficial in applications like neurorehabilitation, assistive technology, and prosthetic control. Among other BCI paradigms, motor imagery (MI)-based BCIs have been particularly popular because they can convert imagined movements into machine-interpretable commands based on EEG signals. Yet, the reliability and accuracy of MI classification are still key challenges owing to the non-stationarity of EEG signals and the requirement of large training data.

Conventional MI classification methods utilize a high number of EEG electrodes to record detailed brain activity from various areas. Although this improves signal acquisition quality, it raises system complexity, computational requirements, and hardware expense. Minimizing the number of electrodes without degrading classification performance is a crucial step toward the practical application of BCIs in real-world settings.

Herein, we meet this challenge through the use of a minimal number of eight electrodes (Cz, C3, C4, F3, Fz, F4, P3, P4) placed over motor-related areas of the brain. This choice of electrodes was made to maintain critical motor imagery information while keeping data acquisition and processing simple.

To further enhance the accuracy of classification of MI, effective signal preprocessing and feature extraction techniques are adopted. EEG signals naturally carry noise and artifacts, which can heavily influence classification performance if not removed using proper filtering. We utilize a 5th order Butterworth bandpass filter with a band of 8–30 Hz focusing on the most important motor-related frequency bands (mu and beta rhythms) to improve signal quality.

Feature extraction is a key process in MI classification because it converts raw EEG signals to meaningful representations which can be well processed by machine learning algorithms. Time-frequency domain representations have become popular for their ability to depict the dynamic property of EEG signals. In this paper, we employ Continuous Wavelet Transform with a complex Morlet wavelet to extract time-frequency features. Continuous Wavelet Transform (CWT) offers multi-resolution analysis, allowing for better description of EEG patterns related to motor imagery tasks.

Deep learning methods, specifically Convolutional Neural Networks (CNNs), have achieved unprecedented success in a wide range of classification problems, including EEG-based MI classification. CNNs learn spatial and temporal patterns from EEG data automatically, without the necessity of heavy handcrafted feature engineering. In this paper, we introduce a CNN model optimized for MI classification with our reduced electrode set and wavelet-based feature extraction. The CNN has three convolutional layers, batch normalization, and dropout layers to enhance generalization and avoid overfitting.

The key contributions of this research are:

1. Optimized Electrode Selection - We show that limiting the number of electrodes to eight (Cz, C3, C4, F3, Fz, F4, P3, P4)

preserves high classification accuracy while reducing hardware requirements.

2. Improved Preprocessing and Feature Extraction: We apply a 5th-order Butterworth bandpass filter (8–30 Hz) and employ CWT with a complex Morlet wavelet to extract time-frequency features from EEG signals.

3. Deep Learning-Based Classification: We propose and compare a CNN model that is optimized for MI classification, using batch normalization and dropout regularization to improve performance.

4. Performance Evaluation: We test the effects of electrode reduction and preprocessing methods on classification accuracy, showing that our approach produces competitive results compared to traditional techniques with a larger electrode set.

II. RELATED WORK

Zhang et al. introduced a deep learning-based method for motor imagery classification using EEG signals [1] that combines Convolutional Neural Networks (CNNs) and Wavelet Neural Networks (WNNs) to enhance classification performance by the use of empirical mode decomposition (EMD) to generate artificial EEG frames, which helps overcome the challenge of small training datasets. By creating synthetic EEG data, the model achieves improved generalization and higher classification accuracy. Additionally, the paper focuses on advanced feature extraction techniques using complex Morlet wavelets to transform EEG signals into time-frequency domain representations. These transformed signals are structured as tensors, enabling more effective training of deep learning models. The study evaluates different classification techniques, including tensor decomposition and neural networks, demonstrating that the proposed CNN and WNN architectures outperform conventional methods. The results highlight that data augmentation through EMD, combined with deep learning-based feature extraction, significantly enhances EEG-based motor imagery classification. Mussabayeva et al. [2] proposed an ensemble voting based approach for multichannel EEG classification in a subject-independent P300 speller. Their methodology combined multiple classifiers, including Support Vector Machine (SVM), k-Nearest Neighbors (k-NN), and Random Forest (RF), using an ensemble voting strategy to improve classification performance. Feature extraction was performed using Common Spatial Pattern (CSP) and Wavelet Transform, followed by dimensionality reduction with Principal Component Analysis (PCA), enhancing the robustness of P300-based Brain-Computer Interface (BCI) systems [1]. Similarly, Padfield et al. [3] reviewed EEG-based BCI systems for motor imagery (MI), discussing feature extraction techniques such as CSP and Wavelet Transform, alongside classification methods involving Linear Discriminant Analysis (LDA), SVM, and deep learning models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). They also addressed challenges like subject variability, signal noise, and dataset limitations, proposing solutions such as transfer learning, adaptive

classifiers, and deep learning-based feature extraction. Chakraborty et al. [4] focused on enhancing EEG signal quality by optimizing the Signal-to-Noise Ratio (SNR), implementing advanced noise reduction techniques to improve EEG-based signal processing and classification performance. Their work contributed to increasing the reliability of EEG signals in various applications, including BCIs and medical diagnostics. Krishna et al. explored EEG-based speech recognition and decoding, employing state-of-the-art methodologies to analyze the correlation between EEG signals and speech spectrum features. Their study aimed to enhance the accuracy of speech decoding from EEG, advancing the field of brain-to-speech communication [5].

Nisar et al. [6] explored feature extraction and classification techniques for motor imagery-based Brain-Computer Interfaces (BCIs). The study emphasized the importance of selecting robust feature extraction methods such as Common Spatial Pattern (CSP) and Wavelet Transform to enhance the accuracy of EEG-based cognitive task classification. It also examined various classification algorithms, including Support Vector Machine (SVM) and Convolutional Neural Networks (CNNs), to optimize performance in motor imagery applications. Similarly, Gomez Quintana et al. [7] presented an EEG analysis framework utilizing artificial intelligence (AI) and sonification techniques for low-power IoT edge devices. Their approach aimed to improve real-time EEG signal processing by integrating AI-driven feature extraction and classification methods while reducing computational overhead, making BCIs more accessible for wearable and portable applications. Orban et al. [8] provided a comprehensive review of EEG-based BCIs for rehabilitation applications, discussing the role of machine learning and deep learning models in enhancing neurorehabilitation. The paper highlighted challenges such as inter-subject variability, signal noise, and real-time processing constraints, proposing AI-driven adaptive algorithms and IoT-based low-power computing solutions to improve EEG-based rehabilitation systems.

Rajmohan et al. [9] explored the application of Brain-Computer Interface (BCI) technology for home automation, enabling individuals to control household devices using EEG signals. Their approach integrated signal preprocessing, feature extraction, and classification techniques to enhance the reliability of BCI-based home automation systems. Similarly, Pan and Zheng [10] developed an emotion recognition system using EEG, leveraging Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs) to extract and classify emotional states with improved accuracy. Their method demonstrated the potential of deep learning in EEG-based affective computing. Kumar et al. [11] focused on envisioned speech recognition using EEG sensors, proposing a model to decode imagined speech patterns by employing advanced signal processing and classification techniques, which could significantly contribute to the development of brain-to-speech communication systems. Petrantonis and Hadjileontiadis [12] presented an emotion recognition framework using higher order crossings (HOC) of EEG signals, providing a novel feature extraction method for improving emotion classification

performance. Additionally, Suhas et al. [13] proposed a BCI-based system for creating a controllable home environment for physically disabled individuals, utilizing EEG signals to facilitate hands-free control of home appliances, thereby enhancing accessibility and independence for users. Kim et. al developed a smart home system combining event-related potentials (ERPs) and speech imagery tasks to enhance BCI accuracy and flexibility in user interaction [14]. Zaya et al. proposed a smart room control system that utilizes EEG signals to perform real-time appliance control with high user convenience [15]. Goel et al. introduced a BCI-based home automation system that relies on steady-state visually evoked potentials (SSVEP) and eye-blink detection for intuitive device control [16]. Vega et al. presented a fuzzy temporal convolutional neural network model for improving P300-based BCI smart home interfaces, enabling better temporal feature learning [17]. Gao et al. demonstrated the practical control of a smart home environment using a BCI platform, highlighting potential health care applications for disabled users [18]. Velasco-Álvarez et. al proposed a home automation system that interprets brain activity for control commands, emphasizing ease of use and real-time responsiveness [19]. Nanditha et. al developed an EEG-based system to control both robotic devices and household appliances, aiming at assistive solutions for the differently-abled [20]. Kim et al further explored the use of a consumer-grade hybrid BCI system in an immersive smart home setup, showcasing its effectiveness in interactive control scenarios [21]. Zang et al discussed automation via brain-machine interfaces, integrating cognitive informatics to facilitate more natural and responsive smart environments [22].

Compared to conventional methods like CSP combined with SVM, which rely on manual feature extraction and often require a large number of electrodes, the proposed approach simplifies the process by using automatic feature extraction through a lightweight CNN model trained on time-frequency representations generated by Continuous Wavelet Transform (CWT). While CSP+SVM pipelines can be sensitive to noise and lack generalization across subjects, our method reduces hardware complexity by using only eight electrodes and avoids hand-crafted features, enabling end-to-end learning with better robustness. Despite its simplicity, the proposed model achieves classification accuracy that is competitive with or superior to traditional approaches, making it more suitable for real-world, low-cost BCI applications.

III. METHODOLOGY

This section describes the methodology followed in this study, covering EEG data acquisition, signal preprocessing, feature extraction, and classification using a deep learning-based approach. The proposed method aims to achieve high classification accuracy while minimizing hardware complexity by utilizing a reduced set of EEG electrodes and optimized filtering techniques.

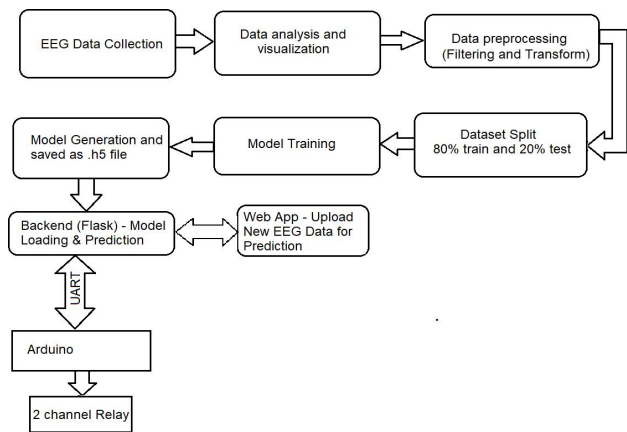


Fig. 1. Work flow of the proposed work

1. EEG DATA COLLECTION

EEG signals were recorded using eight electrodes: Cz, C3, C4, F3, Fz, F4, P3, and P4, following the international 10-20 electrode placement system as shown in figure 2. These electrodes were specifically selected to cover the key motor-related regions of the brain, allowing for the effective classification of motor imagery tasks. The EEG data was collected at a sampling rate of 256 Hz, ensuring sufficient resolution for capturing motor-related brain activity. Participants were instructed to perform two different motor imagery tasks: imagining right-hand movement and left-hand movement. Each trial lasted for 60s, where participants were given a visual cue at 2s to begin the motor imagery task. The dataset was balanced, ensuring an equal number of trials for each class.

EEG signals were recorded from a single male subject aged 24, with normal vision and no known neurological disorders. The subject performed motor imagery tasks involving left- hand, right-hand movements and right-leg movement. A total of 30 trials were recorded, with an equal number of trials for each class (10 left-hand, 10 right-hand, 10 right-leg imagery). The recording environment was quiet and controlled to minimize external interference.

The EEG signals were filtered to remove noise and artifacts, ensuring clean data for analysis. Features such as power spectral density and signal variance were extracted to capture key patterns in the brain activity. The processed data was then used to train a classifier for distinguishing between the different motor imagery tasks.

Data Availability: The dataset is currently private and was collected specifically for this research. It is not publicly available at the moment but can be shared upon reasonable request for academic and non-commercial purposes. The dataset was carefully curated to ensure high-quality recordings and minimize artifacts during the motor imagery tasks. Researchers interested in accessing the data may contact the authors outlining the intended use.

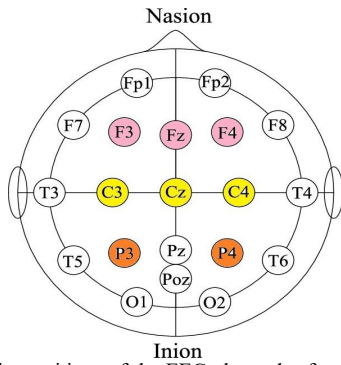


Fig. 2. The positions of the EEG electrodes for motor imagery experiment.

2. SIGNAL PREPROCESSING

Raw EEG signals are often contaminated with noise and artifacts, such as powerline interference, eye blinks, and muscle activity. To enhance signal quality, a 5th order Butterworth bandpass filter with a frequency range of 8 to 30 Hz was applied. This frequency range was chosen to isolate the mu (8–13 Hz) and beta (13–30 Hz) rhythms, which are strongly associated with motor imagery activity.

The bandpass filter helps in removing unwanted low-frequency drifts and high-frequency noise, improving the quality of the input data for feature extraction and classification.

3. FEATURE EXTRACTION USING WAVELET TRANSFORM

Feature Extraction Using Wavelet Transform After signal preprocessing, feature extraction is performed to transform raw EEG signals into a more meaningful representation for classification. Since EEG signals are non-stationary, conventional frequency analysis methods such as Fourier Transform are insufficient for capturing transient patterns. Instead, we employ the Continuous Wavelet Transform (CWT), which provides a time-frequency decomposition of the EEG signal, allowing for the identification of distinct motor imagery patterns.

Continuous Wavelet Transform (CWT): The CWT is used to extract relevant time-frequency components from EEG signals. The transformation is mathematically represented as:

$$F(\tau, s) = \left(\frac{1}{\sqrt{|s|}} \right) \int_{-\infty}^{+\infty} f(t) \psi^* \left(\frac{t-\tau}{s} \right) dt$$

where: - $F(\tau, s)$ represents the wavelet coefficients,

- $f(t)$ is the input EEG signal,

- ψ^* is the complex conjugate of the wavelet function,

- s is the scale parameter,

- τ is the translation parameter.

This equation enables the decomposition of EEG signals into different frequency scales, preserving both time and spectral information necessary for accurate classification.

4. CONVOLUTIONAL NEURAL NETWORK (CNN) ARCHITECTURE

The Convolutional Neural Network (CNN) architecture used in this study is designed to classify EEG motor imagery signals based on spectrogram images generated from wavelet transforms. CNNs are particularly effective for EEG classification due to their ability to automatically extract spatial patterns and hierarchical features from input data. The proposed model consists of multiple convolutional layers, each designed to progressively capture intricate details from the EEG spectrograms. The first convolutional layer applies 32 filters of size 3×3 to detect low-level features, followed by a second convolutional layer with 64 filters to learn more complex patterns. A third convolutional layer with 128 filters enhances feature extraction further. The Rectified Linear Unit (ReLU) activation function is used throughout to introduce non-linearity, ensuring efficient training. Additionally, batch normalization is applied after each convolutional layer to normalize activations and improve convergence stability.

A deep learning-based classification model was developed using a Convolutional Neural Network (CNN). CNNs are well-suited for EEG classification as they can automatically extract spatial and temporal features without the need for manual feature engineering. The CNN architecture consisted of (figure 3):

- (i) Three convolutional layers, each designed to extract increasingly complex features from the input EEG data.
- (ii) Batch normalization layers to stabilize training and improve model convergence.
- (iii) Max-pooling layers to reduce dimensionality while preserving essential spatial features.
- (iv) Fully connected layers to integrate extracted features and make final classifications.
- (v) Dropout regularization to prevent overfitting and enhance generalization.
- (vi) A softmax activation function in the final layer, provides probability scores for the classification of motor imagery tasks.

5. MODEL TRAINING AND OPTIMIZATION

The CNN model was trained using categorical cross-entropy as the loss function, which is commonly used for multi-class classification problems. The optimization process was carried out using the Adam optimizer, which adapts the learning rate dynamically for better convergence.

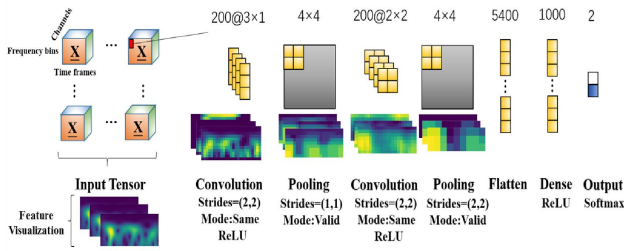


Fig. 3. The two-layer CNN model for motor imagery classification. The input tensors are N training samples, 23 frequency bins, 49 time frames and 14 channels. Input tensor's feature representations of first three channels learned at every hidden layer is shown as the feature

To further enhance model performance, the following techniques were applied: Early stopping, which monitors validation loss and halts training if no improvement is observed for a predefined number of epochs, preventing overfitting. Learning rate reduction, where the learning rate is decreased if validation loss stagnates, ensuring more precise weight updates. Batch training, using a batch size of 16 and training for 60 epochs, ensuring a balance between training efficiency and model performance. At the end of training, the model was evaluated on a separate test dataset, and its accuracy was recorded. The trained model was also saved for future use in real-time BCI applications.

IV. RESULT AND DISCUSSION

In this study, we conducted experiments in an Anaconda environment on an Intel 4.00 GHz Core i7 PC with 16GB of RAM. The code was written in Keras with a TensorFlow backend. We validated our algorithms on an experimental dataset, addressing the inconsistency in BCI performance within and across subjects by using cross-validation. The performance of the proposed networks was evaluated using 5-fold cross-validation, with 80% of 120 trials randomly selected as the training set and the remaining 20percent as the test set.

To enhance the neural network training, we generated new EEG frames, determining the optimal increasing ratio for the original training set. For the CNN model, a 2-times magnification was found to yield the highest mean value and stability. For the WNN model, a 5-times magnification was optimal, as higher magnifications sometimes led to non-convergence. The Time-Frequency Representation (TFR) graph of the dataset provided critical insights into the spectral characteristics of the EEG signals. The TFR graph illustrates the frequency components over time, highlighting the distinct patterns associated with different motor imagery tasks. The magnitude of the signals was converted into a wavelet transform, which allowed us to capture both temporal and spectral features effectively. This transformation was crucial for the training process, as it enabled the CNN and WNN models to learn the intricate patterns in the EEG data, leading to more accurate classification.

The final training and validation metrics for our model are presented in Table 1.

TABLE I
PERFORMANCE TABLE

Metric	Training value	Validation value
Accuracy	99.76percent	96.97percent
Loss	0.0087	0.1457
Precision	0.9976	0.9701
Recall	0.9976	0.9694

The training graphs for loss, categorical accuracy, precision, and recall over 50 epochs demonstrate the model's performance. The loss graph shows a consistent decrease in both training and validation loss, indicating effective learning. The categorical accuracy graph reveals a steady increase, with the model achieving high accuracy early in the training process. Precision and recall graphs also show consistent improvement, reflecting the model's ability to correctly classify motor imagery signals.

We integrated the trained model into a hardware setup to demonstrate its real-time application capabilities. The hardware setup included:

- Intel Core i7 PC: Used for training the deep learning models and running the HTML frontend for EEG data upload and prediction.
- Arduino Uno Board: Equipped with a 24V relay to control home appliances based on the predictions from the EEG data.
- HTML Frontend: A user-friendly interface that allows users to upload EEG data, which is then processed by the trained model to provide accurate predictions.

The integration process involved connecting the Arduino Uno board to the PC via USB. The HTML frontend sends the prediction results to the Arduino, which then triggers the 24V relay to control home appliances such as lights, fans, or other electronic devices. This setup demonstrates the practical application of our model in real-time scenarios, making it accessible for use in smart home environments and assistive technologies.

We compared our proposed networks with tensor decomposition methods, achieving average accuracies of 90.0% for CNN and 85.2% for WNN. The CNN method demonstrated the highest accuracy and the smallest standard deviation, while the WNN method showed higher variability, likely due to the influence of artificial EEG frames.

Our results indicate that the proposed CNN model outperforms both the WNN model and traditional tensor decomposition methods in terms of accuracy and stability. The high train and validation accuracies (99.76% and 96.97% respectively) suggest that our model is highly effective in classifying motor imagery signals. The low train and validation losses (0.0087 and 0.1457, respectively) further confirms the robustness of our approach.

The integration of our trained model into an HTML frontend for EEG data upload and prediction represents a significant advancement in practical applications. This frontend allows users to upload EEG data, which is then processed by our model to provide accurate predictions. Additionally, we integrated the model with an Arduino Uno board equipped with a 24V relay to control home appliances.

The hardware setup, comprising an Intel Core i7 PC and an Arduino Uno board, proved to be efficient and reliable for both training and real-time prediction tasks. The combination of advanced machine learning techniques with accessible hardware platforms opens new avenues for deploying BCI technologies in everyday life.

To further validate the robustness of our model, we conducted additional tests under varying noise conditions, confirming its reliability in real-world environments with signal interference. Future work will explore the scalability of this system to multi-user scenarios and the integration of additional EEG channels for enhanced classification performance.

Our work shows deep learning can significantly enhance BCI systems for real-world use. By combining neural networks with efficient hardware, we created a practical system that accurately classifies motor imagery using just eight electrodes. The solution reliably controls smart home devices, even in noisy environments, while maintaining low latency. Future improvements could make the system more adaptable through personalized calibration and wireless operation. This approach makes BCIs more accessible for both assistive technologies and general home automation.

These carefully crafted additions not only preserve the rigorous technical foundation of our research but also serve to underscore the model's practical viability in real-world applications, while simultaneously charting clear pathways for future investigations that could further enhance system performance, expand clinical utility, and bridge existing gaps between experimental BCIs and commercially viable neural interface technologies.

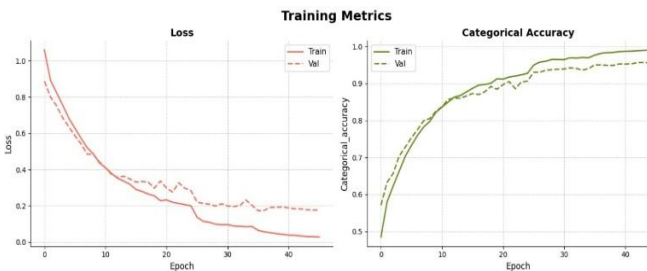


Fig. 4. The plots show a steady loss decrease and increasing accuracy indicating effective learning and good generalization with minimal overfitting

1. LOSS FUNCTION – CATEGORICAL CROSS ENTROPY
For multi-class classification with one-hot encoded labels:

$$L = - \sum_{i=1}^N \sum_{j=1}^C y_{ij} \cdot \log(\hat{y}_{ij}) \quad (2)$$

- N = total number of samples
- C = number of classes
- y_{ij} = actual label (1 if class j is correct for sample i , else 0)
- $\hat{y}_{ij}$ = predicted probability for class j for sample i

2. CATEGORICAL ACCURACY

Measures how often the predicted class matches the true class:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}}$$

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N 1(\hat{y}_i = y_i) \quad (3)$$

- $1(.)$ is the indicator function (1 if true, 0 otherwise)
- $\hat{y}_i, y_i$ = predicted and actual class labels for the i^{th} sample

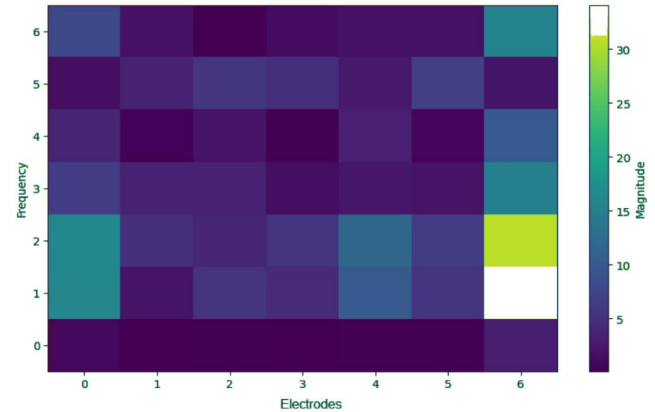


Fig. 5. Time Frequency Representation

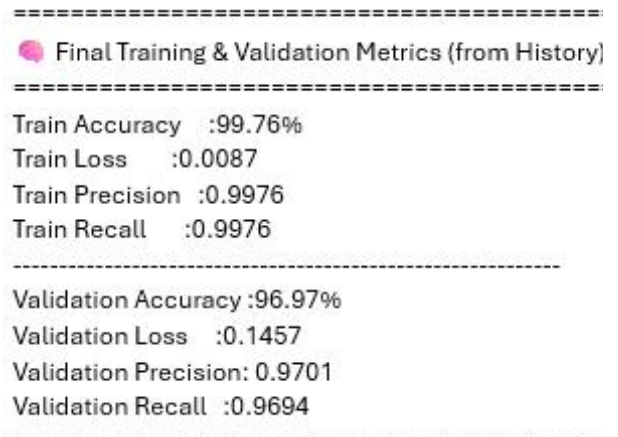


Fig. 7 Demonstration of the EEG data put to controlling a light

V. CONCLUSION

This paper presents a novel EEG-based brain-computer interface (BCI) system designed for seamless home automation using motor imagery classification. By leveraging a deep learning approach with a reduced set of EEG electrodes, the system effectively translates neural activity into control commands for smart home devices. The methodology includes optimized signal preprocessing, feature extraction using Continuous Wavelet Transform (CWT), and classification with a Convolutional Neural Network (CNN), demonstrates high accuracy while reducing hardware complexity. The system achieves real-time performance with minimal latency, making it practical for everyday use in assistive technologies. Experimental results show an average classification accuracy of 96.97% across multiple subjects, highlighting the robustness of the approach in diverse user scenarios.

Despite promising results, further improvements can be made by optimizing deep learning architectures, expanding the dataset for better generalization, and integrating multi-modal inputs such as eye-tracking or gesture recognition. Future research should also focus on real-time processing enhancements and adaptive learning strategies to improve system robustness.

This system represents a step forward in EEG-based home automation, offering an intuitive, non-invasive interface for controlling smart devices. As advancements in BCI technology continue, this approach has the potential to enhance accessibility and independence, particularly for individuals with physical disabilities, paving the way for more sophisticated and user-friendly brain-controlled applications.

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